

PAPER_1079: Galaxy Cluster Cooling-Flow Buoyancy Suppression

Star Magic UQFF Framework — Session 225

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Abstract

We present a parameterized calculator for galaxy-cluster cooling-flow regulation via $F_{U,Bi,i}$ jet-mediated AGN feedback. The model balances intra-cluster medium (ICM) radiative cooling against buoyancy-driven jet heating to predict the cooling-flow suppression factor $S(\Gamma)$ as a function of SCm phonon linewidth.

1. ICM Radiative Cooling

The cooling luminosity of the ICM is dominated by thermal bremsstrahlung:

$$L_{\text{cool}} = n_e^2 \cdot \Lambda(T) \cdot V_{\text{cool}}$$

where the cooling function for metallicity $Z = 0.3$ solar:

$$\Lambda_{\text{ff}}(T) = 1.42 \times 10^{-40} \cdot \sqrt{T} \cdot g_{\text{ff}} \cdot (1 + 1.8Z) \quad [\text{W m}^3]$$

The classical mass deposition rate without AGN feedback:

$$\dot{M}_{\text{cool}} = \frac{L_{\text{cool}} \cdot \mu m_p}{(5/2)k_B T}$$

2. $F_{U,Bi,i}$ Jet Heating

The AGN jet mechanical power is linked to the SMBH through the 26-layer buoyancy force:

$$F_{U,Bi,i} = \sum_{i=1}^{26} [U g_i - U b_i] + U_m + U_A + F_n \cdot S_{26}^{(3)} \cdot \Phi(\Gamma) \cdot E_{\text{net}}$$

The jet heating power:

$$Q_{\text{heat}} = |F_{U,Bi,i}| \cdot \eta_{\text{jet}} \cdot M_{\text{ICM}} \cdot v_{\text{jet}}$$

3. Cooling-Flow Suppression Factor

$$S(\Gamma) = \min \left(\frac{Q_{\text{heat}}(\Gamma)}{L_{\text{cool}}}, 1 \right)$$

The suppressed mass deposition rate:

$$\dot{M}_{\text{suppressed}} = \dot{M}_{\text{classical}} \times (1 - S)$$

When $S \rightarrow 1$: complete cooling-flow quenching (AGN dominates). When $S \rightarrow 0$: no suppression (classical cooling catastrophe).

4. Cluster Applications

Perseus (NGC 1275)

- $T_{\text{ICM}} = 4.0 \text{ keV}$, $n_e = 3 \times 10^4 \text{ m}^{-3}$, $r_{\text{cool}} = 100 \text{ kpc}$
- $M_{\text{BH}} = 3.4 \times 10^8 M_{\odot}$
- $L_{\text{cool}} = 1.98 \times 10^{38} \text{ W}$
- $t_{\text{cool}} = 0.57 \text{ Gyr}$
- $S = 1.0$ (fully suppressed)

Abell 2256

- $T_{\text{ICM}} = 6.5 \text{ keV}$, merging cluster
- $M_{\text{BH}} = 10^9 M_{\odot}$

Virgo (M87)

- $T_{\text{ICM}} = 2.5 \text{ keV}$, relativistic jet ($v_{\text{jet}} = 0.99c$)
- $M_{\text{BH}} = 6.5 \times 10^9 M_{\odot}$

5. Gamma Dependence

The suppression factor varies with phonon linewidth through $\Phi(\Gamma)$ in the $F_{U,Bi,i}$ force. A sweep over $\Gamma \in [0.01, 0.50] \text{ THz}$ shows peak suppression near $\Gamma_0 = 0.10 \text{ THz}$, consistent with the SCm resonance framework.

6. Validation

- All 3 clusters produce valid suppression factors $S \in [0, 1]$
- Cooling times physically realistic (0.01 - 1000 Gyr)
- 10/10 self-tests pass

References

1. Perseus cooling flow: NGC 1275, $M_{\text{cool}} \sim 13 \times 10^9 M_{\odot} \text{ H}_2$
2. Sutherland & Dopita (1993): Cooling functions
3. $F_{U,Bi,i}$ framework: `MAIN_1_CoAnQi.cpp` SOURCE4
4. ICM physics: `gen_muge_ngc1275.py`